

2	(a) (i) <i>either</i> random motion <i>or</i> constant velocity until hits wall/other molecule	B1	[1]
	(ii) (total) volume of molecules is negligible compared to volume of containing vessel <i>or</i> radius/diameter of a molecule is negligible compared to the average intermolecular distance	M1 A1	
		(M1) (A1)	[2]
	(b) <i>either</i> molecule has component of velocity in three directions <i>or</i> $c^2 = c_x^2 + c_y^2 + c_z^2$ random motion and averaging, so $\langle c_x^2 \rangle = \langle c_y^2 \rangle = \langle c_z^2 \rangle$ $\langle c^2 \rangle = 3\langle c_x^2 \rangle$ so, $pV = \frac{1}{3}Nm\langle c^2 \rangle$	M1 M1 A1 A0	[3]
	(c) $\langle c^2 \rangle \propto T$ or $c_{\text{rms}} \propto \sqrt{T}$ temperatures are 300 K and 373 K $c_{\text{rms}} = 580 \text{ ms}^{-1}$ (Do not allow any marks for use of temperature in units of °C instead of K)	C1 C1 A1	[3]

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3 (a) (numerically equal to) quantity of (thermal) energy required to change